On The Gravitational Nature Of Time

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Abstract

We explore a modified set of coordinate transformations that describe time, cosmic inflation and gravitational acceleration as the same process as observed from different reference frames through an unwrapped Kaluza-Klein like geometry. We show that these transformations satisfy Lorentz invariance in a $\mathbf{4+0}$ space-time with a radion scalar potential of $\mathbf{1}$. From this mechanism an equation is derived trigonometrically for galactic orbital velocity, reproducing local observation with only baryonic mass. Through these coordinate transformations, a direct mechanism to the equivalence principle is derived yielding multiple functions producing our peculiar velocity and a secondary velocity, as well as a geometric justification for the discrepency between SNe and CMB relative velocity calculations. These modified coordinate transformations produce a classical notion of simultaneity and indicate a novel model of chronology which resolves multiple temporal pardoxes without the need for selective reference frame bias.

Keywords: gravitation, time, relativity, inflation

1 Introduction

Albert Einstein was insistent on a static Universe during the period in which he wrote his specialized theory of relativity[1–4](SR). What is proposed here is largely a re-examination of the geometry of SR under the assumption that the $\mathbb{R}^3$ space between two arbitrary points elongates as time progresses.

This dilation is then bound to γ, giving a velocity dependent function for $\vec{g}$ and a magnitude that extends into a 4th spatial coordinate, y, similar in construction to an unwrapped Kaluza-Klein (KK) dimension[5, 6]. Through these coordinate transformations time, cosmic inflation and gravitational acceleration become the same process as observed from different reference frames in a space-time geometry where the 4th spatial KK dimension acts as a concurrent density axis.

This geometry gives an alternative model of concurrency and synchronicity as a classical notion of simultaneity is maintained by attributing the additional spatial separation to the 4th spatial dimension, $dy = v\tau\sqrt{\gamma^2 - 1}$ as shown in figure 1. While maintaining classical simultaneity resolves numerous temporal paradoxes without the need for selective reference frame preference, it produces a concurrent axis that describes the ultra-dense state of the early Universe as being separated only by some Δ along this 4th spatial axis.

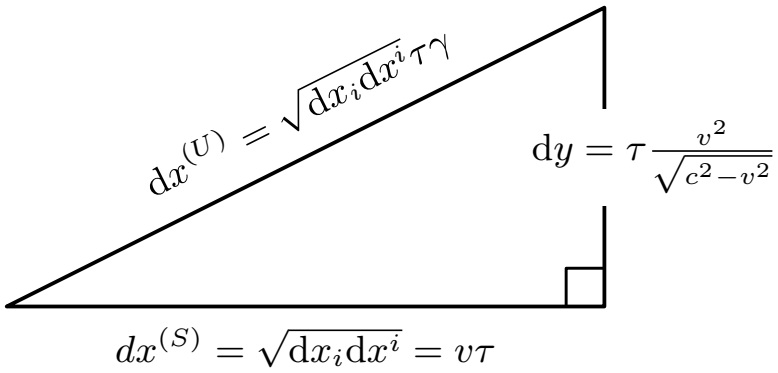


Fig. 1 Right-triangle construction used to represent the ordinary spatial displacement dx, the additional spatial displacement dy, and the extended displacement $dx^{(U)}$. The relation assigned to the hypotenuse is an assumption of the proposed geometric construction.

1.1 The Lighthouse & The Clock-Tower

Let an event be defined classically, such that a single, instantaneous event represents a local measurement as $dt \to 0$.

Let there be two overlapping $\mathbb{R}^3$ coordinate systems, **S** and **U**. Let **S** and **U** represent an equi-temporal surface within a 4-dimensional hyperspace where $y = y^{(S)}$ and $y = y^{(U)}$ in each respective 3-dimensional surface.

This gives

$$f\left(x_i, y\right) = \partial_y \mathcal{H} \tag{1}$$

where

$$f\left(x_i, y^{(A)}\right) = \mathbf{A}\left(x_i\right) \tag{2}$$

for an arbitrary **A** contained within $\mathcal{H}$.

Let **U** represent a 'future' state from the perspective of an observer in **S** so that

$$v^{(S)} = (t, x_\mu) \qquad v^{(U)} = (t + \Delta t, x_\mu) \tag{3}$$

where $\Delta t > 0$ and $\mu = 1, 2, 3, 4$. Likewise, let $\mathbf{S}$ and $\mathbf{U}$ be bound such that

$$f\left(x_i, y^{(U)}\right) = f\left(x_i, y^{(S)}\right) + \int_{x_0^{(S)}}^{x_0^{(U)}} F(x)\ \mathrm{d}t \tag{4}$$

where F is the sum of all forces acting on the point of space at x, integrated according to time, which in turn sums the change between the temporal surfaces $\mathbf{S}$ and $\mathbf{U}$.

Let there also be two observers, A and B. Let A and B agree that B should travel at some *agreed upon* relativistic velocity v' between two arbitrary points in space contained within $\mathcal{H}$, p_1 and p_2. Let the experiment be arranged such that B passes p_1 in $\mathbf{S}$, and p_2 in $\mathbf{U}$.

Let A remain at relative rest with respect to p_1, and at p_2 place a time keeping device large enough to be visible from p_1.

We then make these assertions

1. To maintain a classical model of simultaneity A and B must agree upon the time at which B reaches p_2.
2. To satisfy existing experimental evidence for the additional γ factor[7–11], $\Delta t = \gamma \frac{d}{v}$.

This gives a set of coordinate transformations such that for each relative observer

$$\Delta t = \frac{\overbrace{d \cdot \gamma}^{B}}{v^{(U)}} = \underbrace{\frac{d}{v^{(S)}} = \frac{d}{v^{(U)}\left(\frac{1}{\gamma}\right)}}_{A} \tag{5}$$

2 Theoretical Framework

We acknowledge two primary properties of time:

1. A driving rate of change, at least as fundamental as the $\mathbb{R}^3$ dimensions.
2. A 4th displacement coordinate.

Throughout the following derivations, we presume that time has no physical significance beyond the two properties described above. We presume that time acts only as a mathematical tool: $\Delta t = \frac{d}{v}$. In turn, we must attribute the 4th displacement coordinate to a position along the density axis that arises, y, and the driving rate of change to *motion*.

Likewise, we presume that in a co-moving frame of reference along the y axis, mechanical equations should limit at their Newtonian counterpart with only the relativistic portion of associated magnitudes protruding out of the containing temporal surface.

To remove time as a physically meaningful quantity, we let

$$\tau = \frac{\dot{x}}{\widehat{v}} \tag{6}$$

where $\dot{x}$ indicates not a derivative with respect to time, but rather a fundamental driving rate of change, a magnitude of length that grows with an arbitrary temporal scalar. Let $\widehat{v} = \partial_\tau x \ \hat{\tau}$ where $\hat{\tau} = 1$, or the velocity in units of distance. τ, while equivalent in magnitude to t is derived directly from the geometry and has no explicit time dependence.

2.1 Lorentz Invariance

We consider these coordinate transformations with regards to the Lorentz transformations and find that for an instantaneous event where $dt \to 0$

$$\boxed{\begin{aligned} dx' &= \gamma dx = v^{(U)} \\ c\,dt' &= -\gamma \frac{v}{c} dx = -\partial_\tau y \end{aligned}} \tag{7}$$

where y is as defined in equation 9.
We define $v\,(\gamma)$ to be the Lorentz factor solved for v

$$v\,(\gamma) = c\sqrt{1 - \frac{1}{\gamma^2}} \tag{8}$$

Beginning with the definition of the additional coordinate,

$$y = \tau \frac{v^2}{\sqrt{c^2 - v^2}} \tag{9}$$

we take the partial derivative with respect to τ, holding v constant:

$$\boxed{\partial_\tau y = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{10}$$

Since $x = v\tau$, we can equivalently express the coordinate as a function of x:

$$y = x \frac{v}{\sqrt{c^2 - v^2}}. \tag{11}$$

Taking the partial derivative with respect to x gives

$$\boxed{\partial_x y = \frac{v}{\sqrt{c^2 - v^2}}} \tag{12}$$

We can now express this derivative in terms of the Lorentz factor:

$$\partial_x y = \frac{\gamma v}{c}. \tag{13}$$

Multiplying by the velocity expressed as a function of γ,

$$\begin{aligned} v(\gamma)\partial_x y &= c\sqrt{1 - \frac{1}{\gamma^2}} \frac{\gamma v}{c} \\ &= v\sqrt{\gamma^2 - 1} \\ &= \frac{\gamma v^2}{c}. \end{aligned} \tag{14}$$

Finall we recover

$$\boxed{v(\gamma)\partial_x y = \frac{\gamma v^2}{c} = \partial_\tau y} \tag{15}$$

2.1.1 Invariance of the Minkowski Interval

We can further examine the Lorentz invariance of these transformations by considering the Minkowski interval. For an instantaneous event, where $dt \to 0$, the transformed interval is given by

$$ds'^2 = c^2 dt'^2 - dx'^2. \tag{16}$$

Substituting the coordinate transformations established above,

$$\begin{aligned} ds'^2 &= \left(-\gamma \frac{v}{c} dx\right)^2 - (\gamma dx)^2 \\ &= \gamma^2 dx^2 \left(\frac{v^2}{c^2} - 1\right). \end{aligned} \tag{17}$$

Using the definition of the Lorentz factor,

$$\gamma^2 = \frac{1}{1 - \frac{v^2}{c^2}}, \tag{18}$$

we obtain

$$ds'^2 = -\gamma^2 dx^2 \left(1 - \frac{v^2}{c^2}\right). \tag{19}$$

Therefore,

$$\boxed{ds'^2 = -dx^2 = ds^2} \tag{20}$$

Thus, the instantaneous coordinate transformations preserve the Minkowski interval, demonstrating consistency with the invariance condition of Lorentz transformations.

2.1.2 The Lorentz Factor as a Coordinate Gradient

Having established the relationship between the additional coordinate and the Lorentz transformations, we can rearrange the spatial derivative to express the Lorentz factor directly in terms of $\partial_x y$.

Recall that

$$\partial_x y = \frac{\gamma v}{c}. \tag{21}$$

Squaring both sides gives

$$(\partial_x y)^2 = \frac{\gamma^2 v^2}{c^2}. \tag{22}$$

Using the definition of the Lorentz factor,

$$\gamma^2 = \frac{1}{1 - \frac{v^2}{c^2}}, \tag{23}$$

we can rewrite this expression as

$$(\partial_x y)^2 = \frac{v^2}{c^2 - v^2}. \tag{24}$$

Rearranging and substituting back into the Lorentz factor gives

$$\begin{aligned} \gamma^2 &= 1 + \frac{v^2}{c^2 - v^2} \\ &= 1 + (\partial_x y)^2. \end{aligned} \tag{25}$$

Consequently, we obtain the following expression for the Lorentz factor in terms of the additional coordinate:

$$\boxed{\gamma = \sqrt{1 + (\partial_x y)^2}} \tag{26}$$

This establishes a direct relationship between the Lorentz factor and the spatial gradient of the additional coordinate, allowing the relativistic factor to be expressed geometrically without explicitly introducing velocity.

2.2 The Equivalence Principle

Here we examine these coordinate transformations in the context of the equivalence principle. While a well defined function remains elusive, these generialized coordinate transformations and a numerical intersection with the fine structure constant allows us to derive several important properties of an apparent electromagnetic function producing $\vec{g}$.

2.2.1 Earth's Kinematic Velocity From Electromagnetic Parameters

We define $\delta(M, R)$ to be

$$\delta(M, R) = \frac{d}{dR} g = 2\frac{GM}{R^3} \tag{27}$$

If $\delta(M, R)$ is induced by the coordinate transformations described above, we should presume that this dilation is radial. We then find the magnitude of a single $\mathbb{R}^3$ axis by taking the inverse of the determinant of a diagonal 3×3 matrix:

$$\omega(M, R) = \sqrt[3]{\frac{d}{dR} g} = (\delta(M, R))^{1/3} \tag{28}$$

Investigating this quantity for Earth gives

$$\omega_\oplus = 0.014537\ldots \tag{29}$$

While this approximates 2α, it more closely produces

$$\omega_{\oplus} = \alpha\left(2-\alpha\right) \pm 0.0253\% \tag{30}$$

Then by expanding equation 30

$$\alpha\left(2-\alpha\right) = \sqrt[3]{\frac{d}{dR}g} \tag{31}$$

Moving the electromagnetic parameters and R to one side before integrating with respect to R gives

$$R^4\left(\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6}\right) = 1.272\times 10^{21} \tag{32}$$

We then take the radius independent magnitude, isolating the additional electromagnetic parameters

$$\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{33}$$

As γ represents 1 plus some small ds, let us add 1 to these additional parameters and substitute γ_{em} into equation 8.

$$\gamma_{\text{em}} = 1 + \frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{34}$$

$$v_{(\gamma_{\text{em}})} = 371,721 \text{ m/s} \tag{35}$$

This matches the 2018 Plank data[12] to within 0.514%, and is in rough agreement with general observation of the velocity necessary to induce the CMB dipole.

2.2.2 Solar ω

We expand equation 31 and define $\Gamma\left(\sigma\right)$ to produce

$$\alpha = \Gamma\left(\sigma\right)\left(\frac{d}{dR}g\right)^{1/3} \tag{36}$$

Where $\Gamma\left(\sigma\right)$ is some unknown function of spin. While a well defined function remains elusive, we can take advantage of several characteristics for specific local bodies according to table 1 to derive certain characteristics of σ.

In equation 31 we've demonstrated that for the most mass dense body in the system

$$\Gamma_{\oplus}\left(\sigma\right) \approx \Gamma\left(\alpha\right) = \frac{1}{2-\alpha} \tag{37}$$

Table 1 Advantageous properties

Body	Trait	Value
Earth	The most mass dense.	5495 kg m^{-3}
Saturn	The least mass dense.	687 kg m^{-3}
Sun	The most massive.	1.988×10^{30} kg

We propose that $\Gamma(\sigma)$ acts inside of the determinant. We then take the inverse of the determinant of a diagonal 3×3 matrix

$$\Gamma_{\odot}(\sigma) \propto \sqrt[3]{\Gamma(\sigma)} \tag{38}$$

We then presume that σ is associated with spin. Due to the increased surface gravitational acceleration and the decreased rate of spin for the Sun, we approximate $\sigma = 0$.

$$\Gamma_{\odot}(\sigma) \approx \frac{1}{\sqrt[3]{2}} \tag{39}$$

Or

$$\alpha = \Gamma_{\odot}(\sigma) \sqrt[3]{\frac{d}{dR} g} = \sqrt[3]{\frac{1}{2} \frac{d}{dR} g_{\odot}} \pm 0.47\% \tag{40}$$

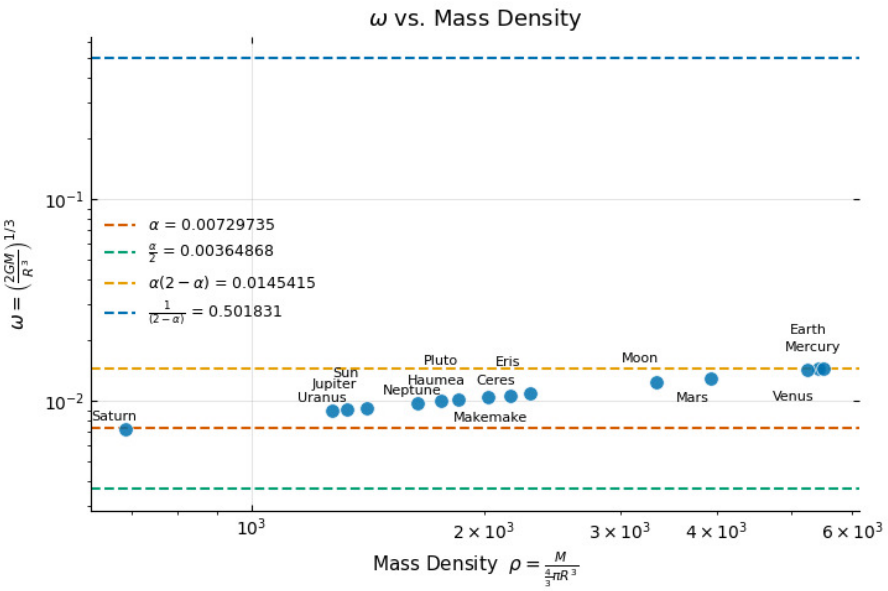


Fig. 2 The model quantity ω, identified with dy and constructed from the inverse determinant associated with volume change, plotted as a function of mass density.

We then find through observation that for the least mass dense body in the system, $\Gamma_{\hbar}(\sigma) \approx 1$.

This gives a set of 3 equations where the $\Gamma(\sigma)$ term cancels for the dominant body in the system.

$$\begin{aligned} g_{\oplus} &= R_{\oplus}\left(\frac{\alpha^3(2-\alpha)^3}{2}\right)\tau \quad && \pm 0.008\% \\ g_{\odot} &= R_{\odot}\alpha^3\tau && \pm 1.426\% \\ g_{\hbar} &= \frac{1}{2}R_{\hbar}\alpha^3\tau && \pm 1.134\% \end{aligned} \tag{41}$$

Which in turn produces a domain where $\alpha \lessapprox \omega \lessapprox \alpha(2-\alpha)$ as shown in figure 2.

2.3 Gravitational Parameters

Due to algebraic symmetries, it's trivial to derive our kinematic velocity through gravitational parameters. What is of interest however, is that we derive what we propose is a more fundamental velocity first. It is only after integrating motion in a manner that is analogous to time that we produce our kinematic velocity. We call this first velocity our *cosmological velocity*, and propose a new geometric interpretation of existing data that may resolve the SNe/CMB velocity discrepancy.

2.3.1 Cosmological Velocity

As γ is applied as a scalar, we take the average of the integral of $\delta_{(M,R)}$ to be

$$\varphi(M,R) = \frac{1}{R}\int \delta(M,R) = 1 + G\frac{M}{R^3} \tag{42}$$

Substituting $\varphi(M,R)$ into equation 8 gives

$$\boxed{v_{\text{cosmo}} = c\sqrt{1 - \frac{1}{\varphi^2(M,R)}} = 525,493.17 \text{ m/s}} \tag{43}$$

While this value aligns closely with the correlation carried out by Gordon et al.[13], there remains large uncertainties inherent in SNe observation[14, 15].

As shown in figure 3, this function only holds for the theoretical upper bounds of mass density. We speculate that there may exist some yet unknown process in the formation of solar systems, at least at our scale and location within the parent galaxy that leads to a natural function of mass density and velocity. While the details of this process have yet to be resolved, the coordinate transformations described here intrinsically link mass density to velocity as $\vec{g}$ becomes a function of v.

2.3.2 Kinematic Velocity

We apply the time-equivalent quantity in equation 6 to equation 42, making equation 42 time dependent. Integrating gives

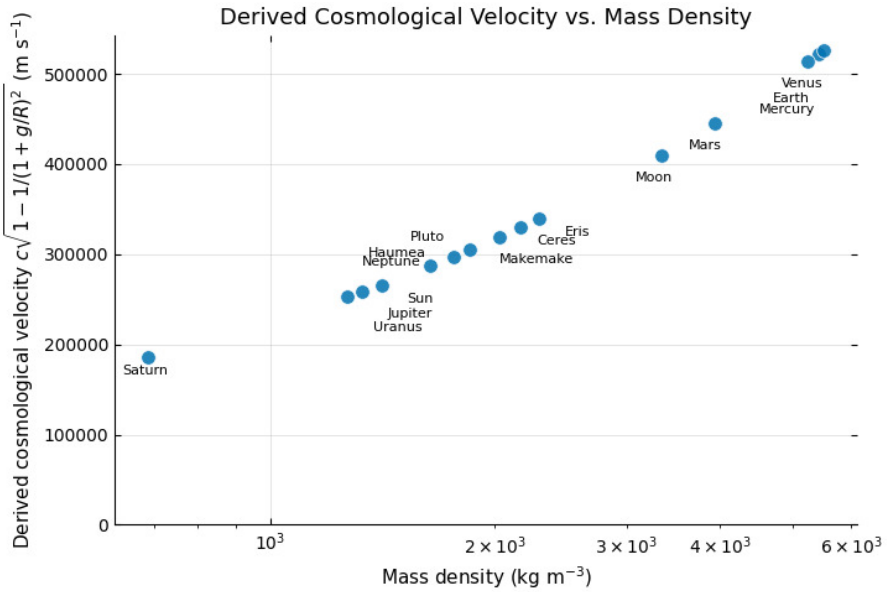


Fig. 3 Model-derived cosmological velocity plotted as a function of mass density.

$$\frac{1}{R}\int_0^R \int_0^x \delta\tau \; dR \; d\tau = 1 + \frac{1}{2}\frac{GM_\oplus}{R_\oplus^3}\left(\frac{\dot{x}}{\hat{x}}\right)^2 \tag{44}$$

We then substitute equation 44 into equation 8 where $d\tau = 1$

$$\boxed{v_{\text{kinematic}} = c\sqrt{1 - \frac{1}{\left(1 + \frac{1}{2}G\frac{M}{R^3}\right)^2}} = 371,580 \text{ m/s}} \tag{45}$$

which matches the 2018 Plank data to within 0.47%.

2.4 Orbital Velocity

2.4.1 Relativistic Stable Orbit Velocity

In the simplified single-body system considered here, the spatial dilation described above permits a trigonometric derivation of the stable-orbit velocity as an alternative to the conventional algebraic equality.

Figure 5 represents the geometric construction used in this derivation.

$$R_0^2 + v_{\text{orbit}}^2 d\tau^2 = \left(\int_0^{R_0} 2G\frac{M}{R^3}\right)^2 = (R_0 + g \; d\tau)^2 \tag{46}$$

Setting $\tau = 1$ gives

$$\boxed{v_{\text{orbit}}\,(M,R) = \frac{\sqrt{GM\,(GM + 2R^3)}}{R^2}} \tag{47}$$

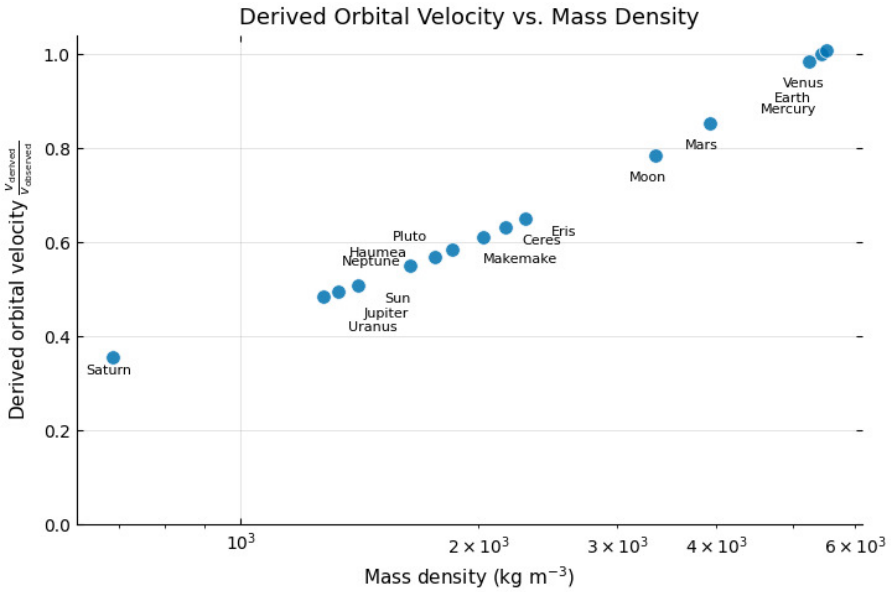


Fig. 4 Model-derived kinematic velocity as a ratio with the velocity found by Eilers et al. plotted as a function of mass density.

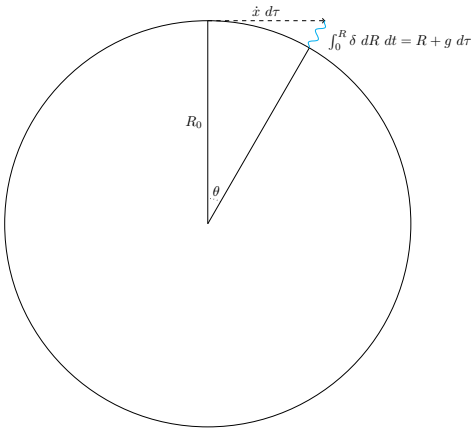


Fig. 5 Geometric construction used in the proposed stable-orbit derivation, showing R_0, the angle θ, the displacement $\dot{x}\,d\tau$, and the proposed radial increment $R + g\,d\tau$.

For the comparison with the Milky Way, we adopt the enclosed Galactic baryonic mass reported by Cautun et al. [16]:

$$M_{\rm enc} = M_{\rm bar}(< R_0) = 4.44 \times 10^{10}\, M_\odot, \tag{48}$$

Here, $R_0 = 8.12\,\mathrm{kpc}$ denotes the Sun's Galactocentric orbital radius[17, 18].

For comparison with observation, we use the solar-neighborhood circular velocity reported by Eilers et al. [19]. Using the adopted enclosed mass and solar orbital radius, the model gives the following orbital velocity:

$$v_{\text{orbit}}\left(M_{\text{enc}}, R_{\text{orbit}}\right) = 216,874 \text{ m/s} = v_{\text{eilers}} \pm 5.59\% \tag{49}$$

The model velocity equals the comparison velocity at $M_{\text{enc}} \approx 4.95 \times 10^{10} M_{\odot}$.

The adopted mass scale is also consistent with the characteristic stellar mass scale of approximately $5 \times 10^{10}\, M_{\odot}$ identified by Yu et al. [20] in their FIRE-2 simulations of Milky Way-mass galaxies. Yu et al. associate this scale with a transition in galactic morphology from predominantly disc-dominated to increasingly spheroid-dominated systems.

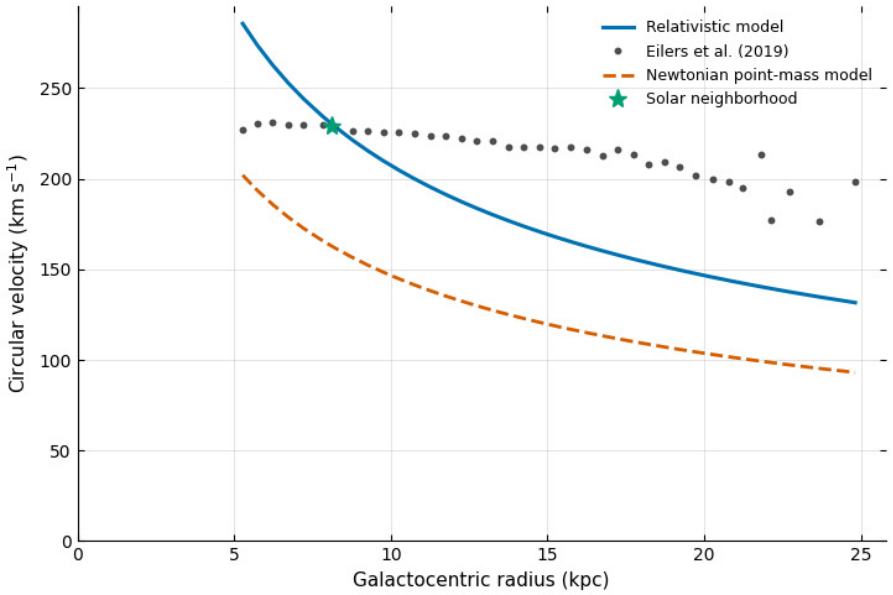


Fig. 6 Milky Way circular velocity as a function of galactocentric radius. Eilers et al. (2019) measurements are shown as points, alongside the relativistic and Newtonian point-mass model curves. The solar-neighborhood measurement is marked with a star; error bars indicate the reported uncertainties in galactocentric radius and circular velocity.

It is important to note here that $v\left(M, R\right) \approx v_{\text{escape}}$ at $R = R_{\oplus}$ and $M = M_{\oplus}$, not a stable orbit velocity. This discrepency needs to be accounted for by Machian resistance in an n-body system. The associated equations remain elusive, but geometric unity achieved elsewhere appears to indicate the viability of this geometry, such that the sum of this radiating force for all other bodies in the system may induce inertial resistance.

2.4.2 Kinematic Velocity From Triginometric Orbit Equations

We now investigate equation 49 by setting

$$v_{\text{orbit}} = \sqrt{2\frac{GM}{R}} \tag{50}$$

which is obtained by $E_{\text{kinetic}} = \int g \, dR$. Solving equation 46 for τ and substituting v_{orbit} gives

$$\mathrm{d}t = \frac{2R^3}{GM - 2R^3} = -1.000000768 \tag{51}$$

Inserting this quantity into equation 8 gives

$$v_{(|\mathrm{d}t|)} = 371,580 \text{ m/s} \tag{52}$$

which matches the 2018 Plank data to within 0.47%.

2.5 Fifth-Dimensional Spatial Displacement and Gravitational Force

While most of the derivations to this point have relied on simplified single body solutions, we explore the connections to the KK 5-dimensional metric here. As the KK model imposes a cylindrical condition largely due to the lack of an observable 4th spatial dimension, we derive the magnitude of this 4th dimension from the coordinate transformations described above and apply no such condition.

2.5.1 Geometric Construction of the Fifth-Dimensional Displacement

We extend the ordinary three-dimensional displacement into the additional spatial coordinate y. Let dx denote the magnitude of the ordinary spatial displacement,

$$dx = \sqrt{dx_i dx^i}, \qquad i = 1, 2, 3. \tag{53}$$

For spatial velocity magnitude v and displacement parameter τ, as defined in the theoretical framework, the ordinary displacement is

$$dx = v\tau. \tag{54}$$

The proposed construction treats the extended displacement $dx^{(U)}$ as the hypotenuse of a right triangle whose perpendicular components are dx and dy as shown in figure 1. With γ denoting the Lorentz factor defined in Equation 8,

$$dx^{(U)} = v\tau\gamma. \tag{55}$$

Applying the Pythagorean theorem,

$$\left(dx^{(U)}\right)^2 = dx^2 + dy^2, \tag{56}$$

substitution gives

$$dy^2 = (v\tau\gamma)^2 - (v\tau)^2 \tag{57}$$

$$= v^2\tau^2\left(\gamma^2 - 1\right). \tag{58}$$

Taking the positive magnitude of the additional displacement yields

$$dy = v\tau\sqrt{\gamma^2 - 1} = \tau\frac{v^2}{\sqrt{c^2 - v^2}}. \tag{59}$$

The velocity form follows from the Lorentz-factor relation in Equation 8.

For constant v, differentiation with respect to τ gives the rate of displacement along y:

$$\boxed{\partial_\tau y = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{60}$$

2.5.2 Incorporation into the Kaluza–Klein Metric

We represent the additional spatial coordinate through the metric component G_{44}. For a diagonal five-dimensional metric, the line element is

$$dS^2 = g_{\mu\nu}dx^\mu dx^\nu + G_{44}dy^2. \tag{61}$$

The corresponding metric tensor has the block structure

$$\boxed{G_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & G_{44} \end{pmatrix}} \tag{62}$$

The component G_{44} sets the metric scaling of the additional coordinate. Its radial variation determines the coupling between motion along y and radial motion in the ordinary spatial coordinates.

For radial coordinate R, the Christoffel symbol associated with this coupling is

$$\Gamma^R_{44} = -\frac{1}{2}g^{RR}\frac{\partial G_{44}}{\partial R}. \tag{63}$$

For a normalized radial coordinate, $g^{RR} = 1$, so

$$\boxed{\Gamma^R_{44} = -\frac{1}{2}\frac{\partial G_{44}}{\partial R}} \tag{64}$$

2.5.3 Finding G_{44}

We take

$$\partial_x y = \frac{v}{\sqrt{c^2 v^2}} \tag{65}$$

Or

$$dy = \frac{v}{c^2 - v^2}dx \tag{66}$$

We then take the KK line element to be

$$dS^2 = g_{\mu\nu}dx^\mu dx^\nu + G_{44}dy^2 \tag{67}$$

and the $\mathbb{R}^4$ spatial separation to be

$$dS^2_{\text{spatial}} = dx^2 + dy^2 \tag{68}$$

where $dx^2 = dx_i dx^i$ for $i = 1, 2, 3$. Expanding this gives

$$dS^2_{\text{spatial}} = dx^2 \left(1 + \frac{v^2}{c^2 - v^2}\right) = dx^2\gamma^2 \tag{69}$$

which matches the expected displacement magnitude of the Lorentz transformations.

Applying a flat metric to the line element defined in equation 67 gives

$$dS^2_U = dx^2 + G_{44}dy^2 \tag{70}$$

yielding

$$\boxed{G^{(U)}_{44} = 1} \tag{71}$$

for $dS^2_U = \gamma^2 dx^2$.

However solving equation 70 for G_{44} in the **S** coordinate system where $dS^2_S = d_i d^i$ gives

$$\boxed{G^{(S)}_{44}dy^2 = 0} \tag{72}$$

indicating that Newtonian mechanics remains contained within a temporal surface, while the relativistic portion of displacement magnitude extends into the 4th spatial dimension, y. For this reason, we speculate that inertia has it's origins purely in the relativistic component of displacement; a consequence of a Machian-like resistance along the 4th dimension.

3 Discussion

3.1 The Appearance of Electromagnetic Parameters

We do not claim to derive a direct mechanical derivation for the fine structure constant, nor an explanation for it's appearance in section 2.2. We simply apply the coordinate transformations defined in section 2 and discover that α appears repeatedly. While the probability is difficult to quantize, we feel that this repeated appearance of α, appearing in gravitational approximations for all 3 of the local bodies with boundary like characteristics likely indicates a shared function which itself contains or produces α. Due to the hypersphere-like geometry that arises from this model, we propose further investigation into the boundary between Wyler's derivation[21] of α and the $4 + 0$ space-time described here.

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